

Metrics One-Pager — ITDX26

Preregistered targets, timing budgets, reproducibility discipline

Timing budgets (live, on-site)

Metric	Target
Cold start (both laptops)	≤ 30 s
Task 12 live inference	≤ 90 s
Task 13 live inference	≤ 120 s
Task 8 MYCA COA convergence	≤ 240 s
Task 14 Map refresh	≤ 5 s per update
Integrated Task 12+13+8 end-to-end	≤ 420 s
DIRTNet reconciliation after restored contact	≤ 300 s
Memory gate (auto-downgrade)	80% VRAM ceiling

Reproducibility discipline

- Seeds: {11, 17, 23, 29, 31}.
- Grouped bootstrap by NLM-Bench dataset family.
- Paired significance tests within seed.
- Held-out data splits documented in every run manifest.
- Merkle root produced per run; chained across the 8-day event.

Task 12 headline metrics

- Precision / recall / F1 with 95% CI.
- Time-to-detection (median + 95% CI).
- Expected Calibration Error, Brier score.
- Class-stability across seeds.
- Conformal interval width at $\alpha = 0.1$.

Task 13 headline metrics

- Held-out link-prediction AUROC + AUPRC.
- Path recall at k for evidence-path queries.
- Community-detection ARI vs ground truth.
- Counterfactual-consistency after node removal.

Task 8 headline metrics

- Time to converge (bounded ≤ 240 s).
- Action-envelope completeness.
- Unsupported-assertion rate (headline: MYCA vs single-agent).

- Safety-policy compliance rate.
- SME 5-point rating.

Task 14 headline metrics

- Cold start ≤ 30 s.
- Uncertainty-envelope coverage vs empirical error.
- SME usability 5-point rating.
- Fallback-state label correctness (100% required).

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